

QUALITY DEFECT LIBRARY CART

AMPHENOL INDUSTRIAL OPERATIONS - DESIGN ENGINEERING INTERN PROJECT

MOBILE STORAGE AND DISPLAY SYSTEM

During my internship, I led the development of a mobile quality-defect library to store and display physical examples of manufacturing defects. The system needed to protect the samples from dust, keep them organized, and allow technicians to remove individual parts for inspection without disturbing the rest of the tray.

The completed system combined a mobile cart with covered tray slots, modular trays, and removable part-retention modules. The trays fit inside a standard Uline box and cart, allowing the library to be reorganized or expanded as new defect samples were added.

PART-RETENTION MODULE

The part retention module consists of two components:

- A rigid outer housing that connects the module to the tray
- A flexible inner insert that grips and supports a Radsert contact

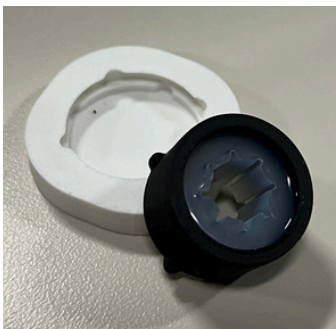
Separating the module into two parts allowed the tray connection and part-retention geometry to be developed independently and quickly. The housing securely connects the module to the tray without tools, while the insert can be adjusted for different part sizes and retention requirements.



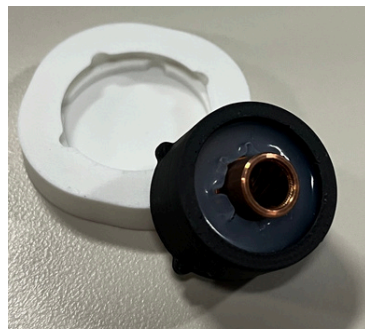
Completed Quality Defect Library Cart



Open Tray of Completed Cart



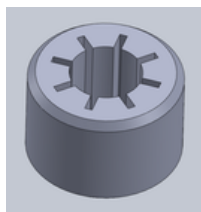
Silicone Module



Silicone Module with Radsert



Exterior Housing CAD



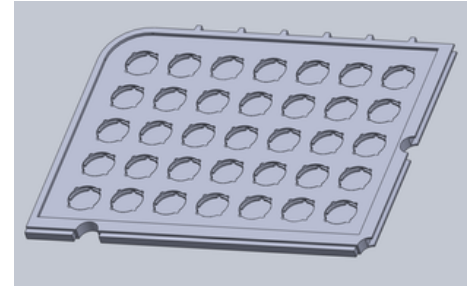
Retention Insert CAD

QUALITY DEFECT LIBRARY CART

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QUICK-RELEASE TRAY CONNECTION

Each module is retained by a three-level rotating locking mechanism. The module is inserted, rotated 45°, and lowered into a slot that prevents it from rotating or falling out. This tool-free connection allows modules to be quickly removed, replaced, or rearranged.

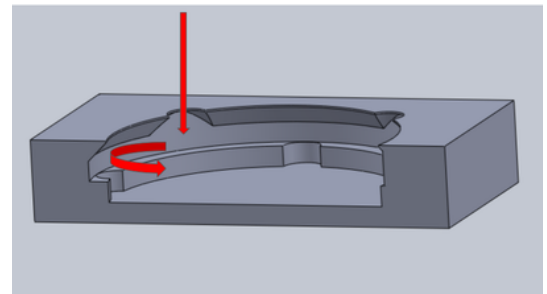


Modular Tray Piece CAD

DEVELOPING THE RADSOK RETENTION GEOMETRY

The flexible insert supports the Radsert on a small rib or ledge inside a ribbed cylindrical opening. The geometry was required to prevent the part from falling out while keeping the removal force low enough for regular inspection.

I created several insert designs with different opening and rib geometries and tested their pullout forces using equipment in the Amphenol test lab. I selected the design with the lowest practical pullout force that still retained the part when the module was dropped.

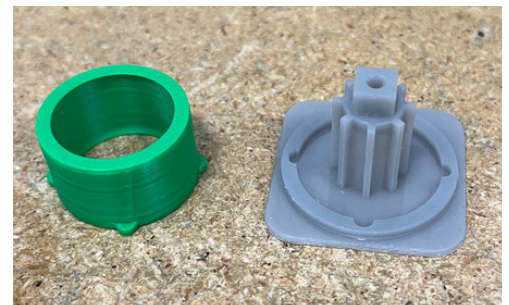


Section View of Locking Mechanism

SILICONE MANUFACTURING PROCESS

The first inserts were made from 40A silicone. I mixed and poured the silicone, placed it in a vacuum chamber to remove trapped air, and cured it in an oven at 60°C.

To save several hours of 3D printing time, I printed the interior mold in resin and allowed the silicone to cure directly inside the outer housing. However, the SLA-printed resin interfered with the silicone curing process, reducing part quality.



Silicone Mold Interior & Outer Housing

TRANSITION TO TPU

I transitioned the insert to FDM-printed 65A TPU because the silicone production process was time-consuming and produced lower-quality parts than expected. Printing directly in TPU simplified manufacturing by eliminating the molding, vacuum degassing, and curing steps while still providing the flexibility needed to retain the Radsert contact.



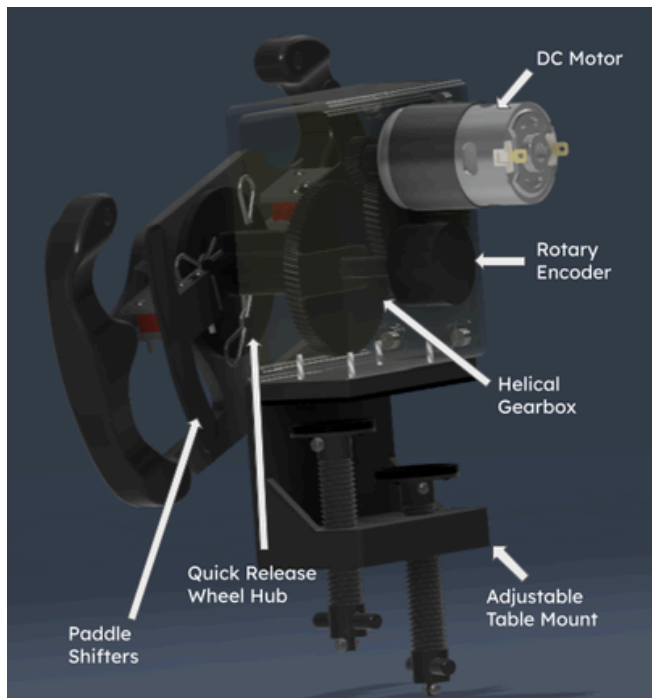
TPU Module with RADSOK

3D-Printed Force Feedback Racing Wheel

INDEPENDENT PROJECT

Designed and built a low-cost force-feedback racing wheel that is easy to manufacture, assemble, and transport. The system uses mostly 3D-printed components, a compact table-mounted design, and removable connections for portability.

CAD DESIGN



Fusion 360 model of the wheel assembly showing the motor, rotary encoder, helical gearbox, paddle shifters, quick-release hub, and adjustable table mount.

FINAL PROTOTYPE



Completed physical prototype

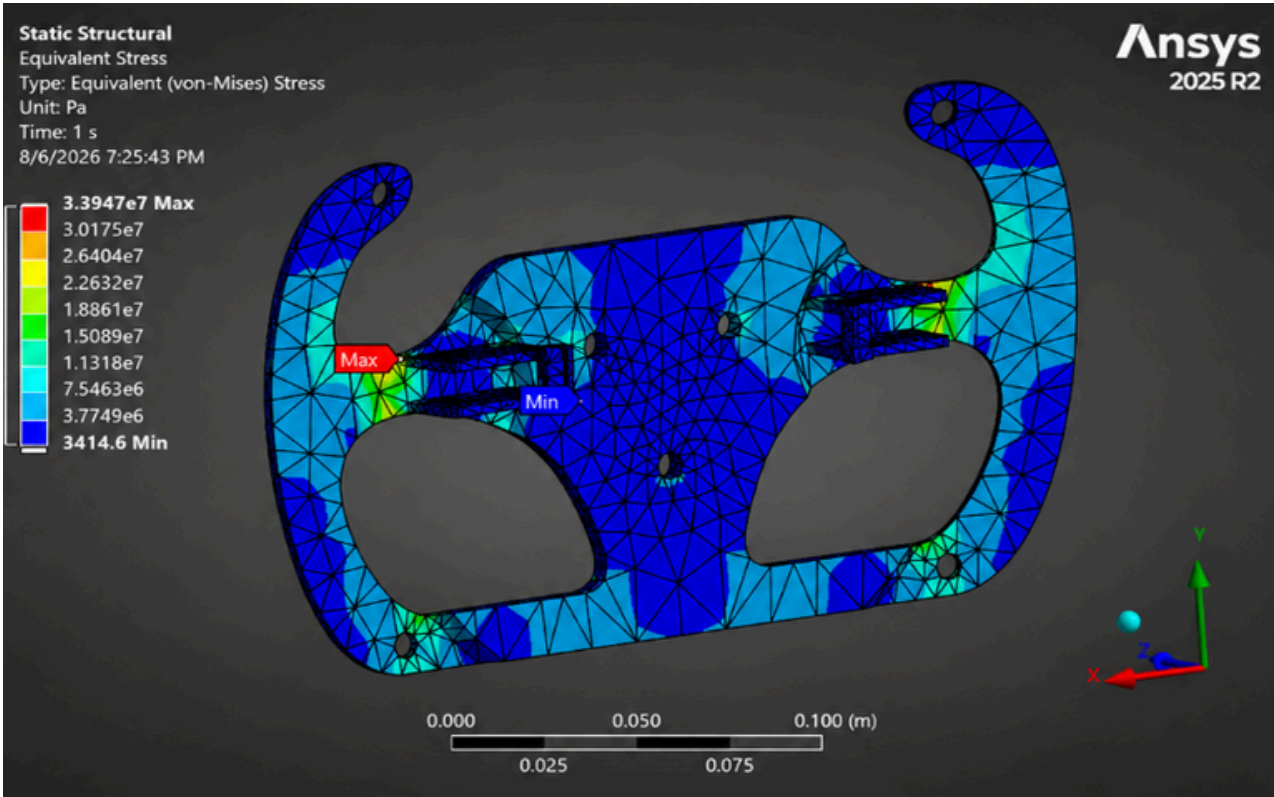
DESIGN REQUIREMENTS & CHALLENGES

I defined the primary design requirements as low cost, portability, ease of manufacturing, and ease of assembly. To meet these goals, I selected 3D-printed PLA because it is affordable, widely available, and allows parts to be produced quickly without specialized tooling. The assembly was designed with accessible fasteners, modular components, and removable electrical connections to simplify construction, maintenance, and transportation. The main challenge was creating a compact printed structure with enough stiffness and strength to support the motor, gearbox, and steering loads without excessive flex or misalignment.

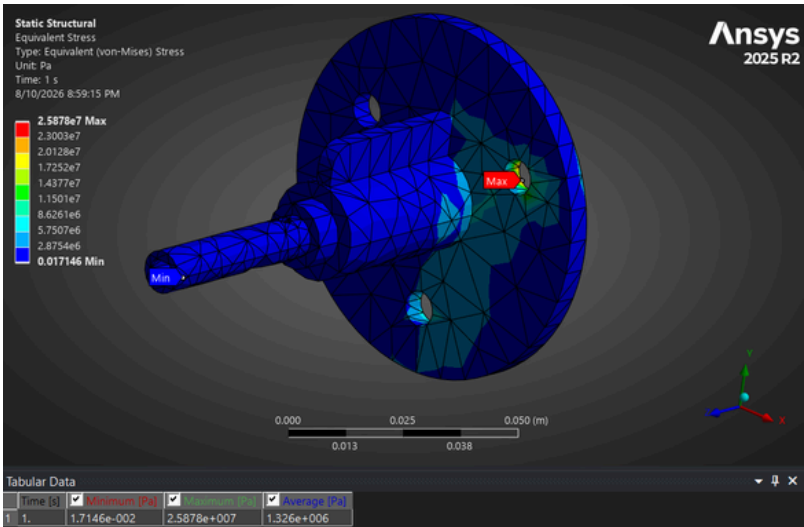
3D-Printed Force Feedback Racing Wheel

INDEPENDENT PROJECT

FINITE ELEMENT ANALYSIS



An 80 N force was applied to each handle, simulating a human using the wheel with excessive force. The analysis showed a maximum von Mises stress of 33.95 MPa and an estimated factor of safety of approximately 1.15–1.39 meaning the wheel will not fail under excessive load.



A 10 N·m moment was applied to the face of the wheel hub, simulating an accidental twisting motion during use of the simulator wheel. The analysis showed a maximum von Mises stress of 25.88 MPa and an estimated factor of safety of approximately 1.81, indicating the wheel should withstand the applied load without failure.